

Introduction to Image Processing Final Project Proposal

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Problem Description

There are many ways to implement image resizing. Content-aware seam carving is one of them. It gained the distribution of the energy of the image, and decided the path of a single seam that doesn't cut through important content. Removal of seams makes the resized image look more natural than just neighbor sampling or interpolation.

An experiment tried to improve seam carving by preprocessing the image with an edge-aware smoothing-sharpening guided filter. The filter is smoothing or sharpening depends on the parameter tuned. By applying the filter, it smooths contents that are usually taken as less important without losing the edge. Thus, it is helpful to decide the path of seam.

Methods

For an input image, first, denoise image to avoid its distraction. Then, applied the guided smoothing filter to the image to smooth the background but preserve edge at the same time. Then calculate the energy map of the image, finally, design the seam carving dynamic programming algorithm to perform image resizing.

Possible results

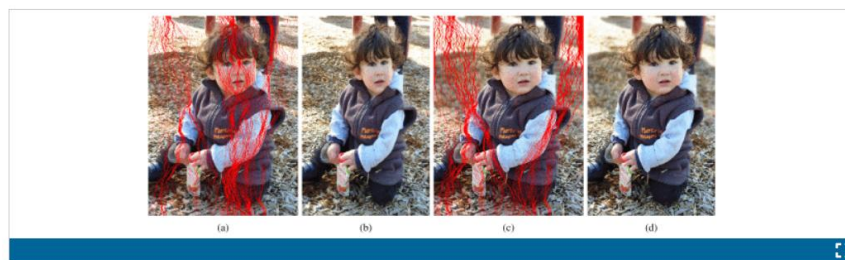


FIGURE 12. Seam carving results. (a) Original image and seams to be removed. (b) Removed seams in (a). (c) Image after background smoothing and seam to be removed. (d) Removed seams in (c). The proposed background filter leads to a better result.

References

G. Deng, F. Galetto, M. Al-nasrawi and W. Waheed, "A Guided Edge-Aware Smoothing-Sharpening Filter Based on Patch Interpolation Model and Generalized Gamma Distribution," in *IEEE Open Journal of Signal Processing*, vol. 2, pp. 119-135, 2021,